

Adya Srivastava

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EDUCATION

BITS Pilani

B.E. Electrical and Electronics Engineering

Pilani, Rajasthan

Aug. 2022 – May 2026

Delhi Public School Ghaziabad Vasundhara

XII Science – 93.1% | X General – 97%

Ghaziabad, UP

2019 – 2021

WORK EXPERIENCE

ML Engineering Intern

Nurix AI

Jul. 2025 – Dec. 2025

- Built **TinyBERT+NN voicemail detection** with **backend integration and model deployment**, achieving **95% precision**.
- Optimised **transformer-based STT pipeline** via **semantic parsing**, improving real-time turn-end detection accuracy by **115%**.
- Designed **WER/EER correction pipeline** using **Word2Vec**, phonetic normalisation, and LLM scoring, reducing WER by **40%**.

Quant Investment Analyst

FidelFolio Investments

May 2025 – Jun. 2025

- Built **quantitative research workflow** for portfolio analytics and Sharpe evaluation, improving research efficiency by **90%**.
- Optimised **risk-aware portfolio construction** via genetic algorithms, and bayesian optimisation, improving Sharpe by **~42%**.
- Analysed multifactor portfolio interactions across **24 factors and 200+ portfolios** for strategy evaluation and signal analysis.

Research Engineer

Grasim (Aditya Birla Group)

May 2024 – Jul. 2024

- Designed **20+ PLC-based railway simulations** for autonomous control, operational validation, and safety compliance.
- Engineered **IR + IoT sensor networks** for real-time railway-track monitoring, achieving **98%** trespassing detection accuracy.

Product Development Lead

Khageshvara Aviation Technology

May 2023 – Oct. 2023

- Developed **ML-CNN aerial vision pipelines** for real-time human detection and accelerated UAV deployment via partnerships.
- Built **investor-grade financial models & pitch strategies**, securing **DST funding** and supporting product development.

PUBLICATIONS

Diagnosing Structural Fragility in LLM Reasoning | [CAISc 2026](#) | [Paper](#) | [Repository](#)

- Built a **multi-probe method** spanning rename, session coupling, and contamination exposure to probe **retrieval vs. reasoning**.
- Ran **adversarial perturbation tests** across **5 frontier LLMs** with **statistical significance testing** to audit robustness.

PROJECTS

FraudScope360 – Multimodal Fraud Detection | [FastAPI](#), [Graph ML](#), [XGBoost](#)

- National Winner, Citi Bank Campus Innovation Challenge**: built a 5-module fraud engine using Graph ML and XGBoost.
- Implemented graph embedding, anomaly detection, and feature engineering pipelines for fraud-ring and identity fraud detection.

Diffusion-Generated Volatility Surfaces for RL Hedging | [DDPM](#), [PPO](#), [PyTorch](#), [CVaR](#)

- Built NIFTY 50 IV surfaces from **912K+** option rows and trained **DDPMs**, achieving 0.005 arbitrage penalty vs. market 0.009.
- Trained PPO hedging agent on DDPM-generated surfaces, reducing **CVaR(95%) hedge error by 60.1%** vs. Black-Scholes.

POSITIONS OF RESPONSIBILITY

Core Member, Department of Sponsorship & Marketing

Independent Student Body, Campus Events – BITS Pilani

- Directed sponsorship acquisition within a 25-member team, securing **Rs. 67+ lakhs** for 2 flagship campus fests.
- Closed sponsorship deals with Atlassian, Walmart, Eightfold AI, and Cisco, generating **Rs. 28+ lakhs** through targeted outreach.
- Built risk-based Excel supply model allocating kind sponsorships to **10,000+ participants** across 35+ sponsor partners.

Core Member, ML Team – Association for Computing Machinery

Independent Student Body, CS Projects – BITS Pilani

- Engineered **OmniMesh**, an offline disaster-response mesh network with edge AI inference, Firebase sync, and P2P routing.
- Built **FraudSense AI** to detect LLM-assisted interview fraud using DetectGPT, biometrics, and LLM-judge fusion scoring.
- Engineered **Data in Motion**, a MILP + Kafka cloud optimisation system for intelligent storage orchestration and migration.

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, Sklearn, XGBoost, LightGBM, Transformers, DDPM, PPO, LangChain, OpenCV

Quantitative Finance: Portfolio Optimisation, Volatility Modelling, Factor Models, Backtesting, CVaR, Time-Series ML

Infrastructure: FastAPI, Docker, Kubernetes, AWS, GCP, Kafka, Redis, MongoDB, Firebase, Linux

Languages: Python, C++, Java, SQL, TypeScript